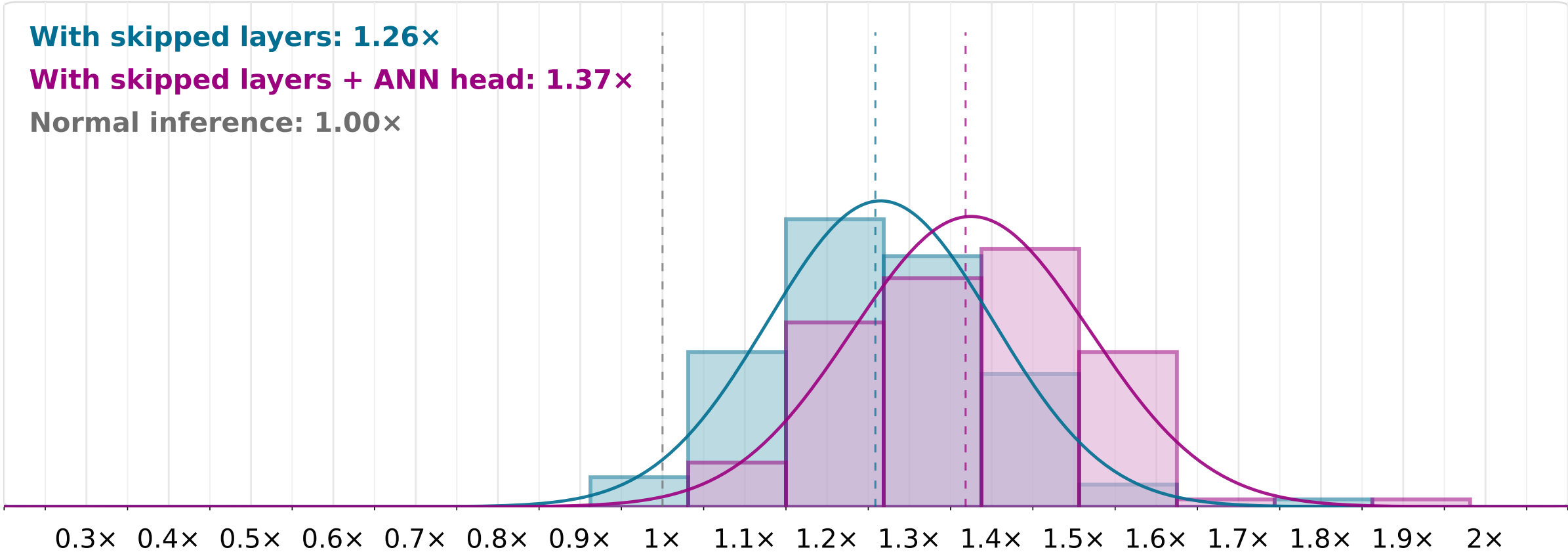


Self-speculation per-token speedup

Qwen3 4B Instruct 2507



Setup

Prompt set = **concrete**
Gap = **(1, 1)**
Block size = **2**
Layers = **36**
LM vocab = **151,936**
LM params = **4.02B**
Head params = **389.0M**
Head portion = **9.67%**

Runtime

GPU = **NVIDIA L4**
Backend = **sdpa**
Model dtype = **bfloat16**
Bridge dtype = **bfloat16**
Speed internals = **no**
Profile prompts = **15**
Warmup prompts = **5**
Measured prompts = **120**

Results

Peak mem normal = **7.61 GiB**
Mean speedup (w/o ANNH) = **1.26×**
Acceptance rate (w/o ANNH) = **36.2%**
Exact match (w/o ANNH) = **48.3%**
Peak mem self (w/o ANNH) = **7.62 GiB**
Mean speedup (w ANNH) = **1.37×**
Acceptance rate (w ANNH) = **35.9%**
Exact match (w ANNH) = **48.3%**
Peak mem self (w ANNH) = **7.67 GiB**
Tokens gen = **s 22,080 / n 21,723**

Profile

Drafter split (w/o ANNH) = **B 42.4% / H 48.7% / O 8.9%**
Verifier/normal (w/o ANNH) = **1.06×**
Drafter/normal (w/o ANNH) = **16.3%**
Drafter split (w ANNH) = **B 72.5% / H 11.9% / O 15.6%**
Verifier/normal (w ANNH) = **1.06×**
Drafter/normal (w ANNH) = **9.7%**
ANNH accuracy = **99.1%**
ANNH head speedup = **6.86×**
ANNH index = **9496 clusters; top-k 100**